Database Systems

Lecture #03

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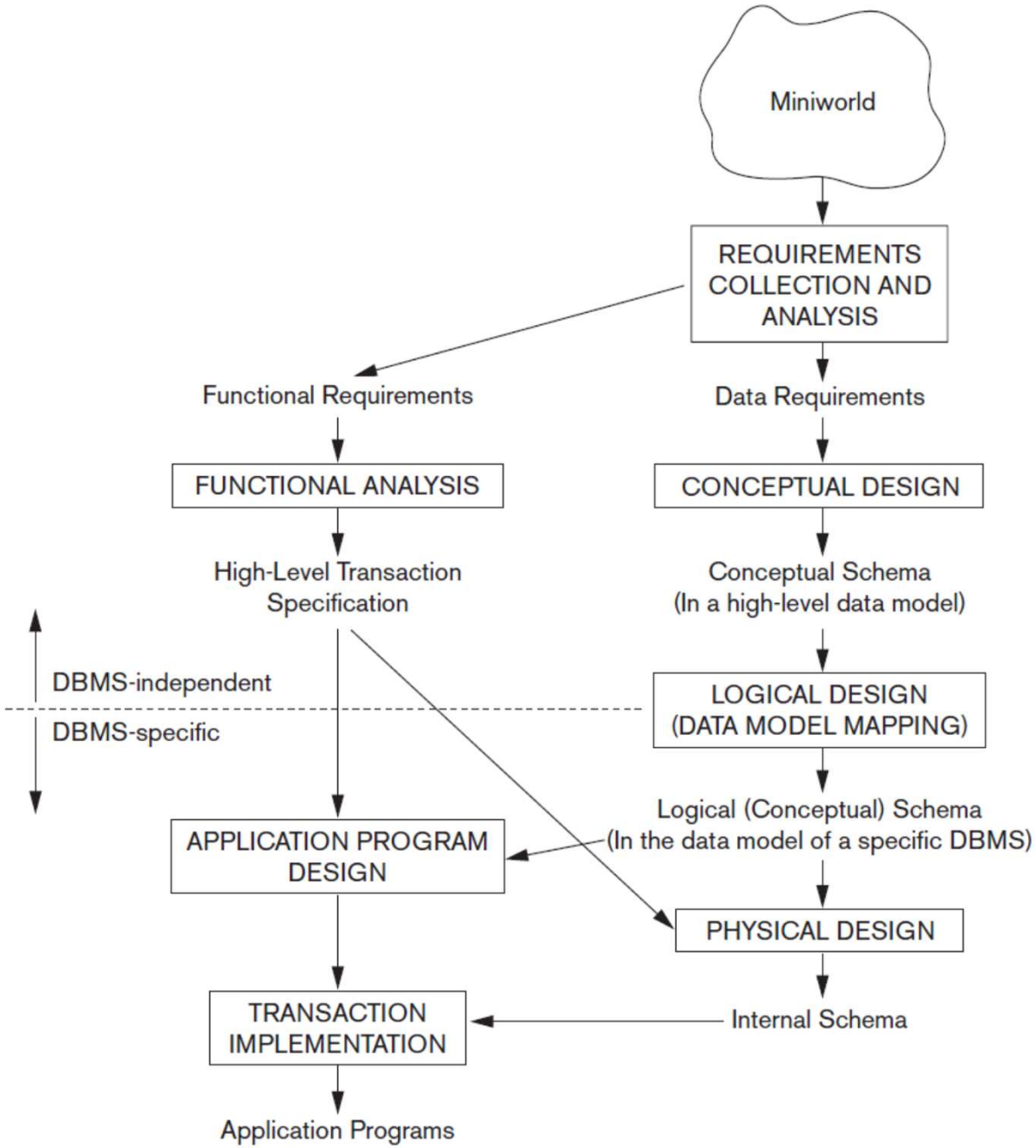
Objectives

- To learn conceptual modeling using the entity-relationship (ER) model
  - Database design process
  - Entity-relationship model
  - Conceptual data modeling

Outline

- Database Design Process
- Example: COMPANY Database
- Entities and Attributes
  - Types of Attributes
  - Entity Type
  - Key Attribute
- Initial Conceptual Design of COMPANY Database
- Relationship
  - Relationship Degree
  - Recursive Relationships

Database Design Process



COMPANY Database

- Requirements
  - Employees, departments, and projects
  - Company is organized into departments
  - Department: has unique name, unique number, locations, manager and start date when manager began managing the department

COMPANY Database

- Requirements (cont'd)
  - Department controls a number of projects
  - Project: has unique name, unique number, and location

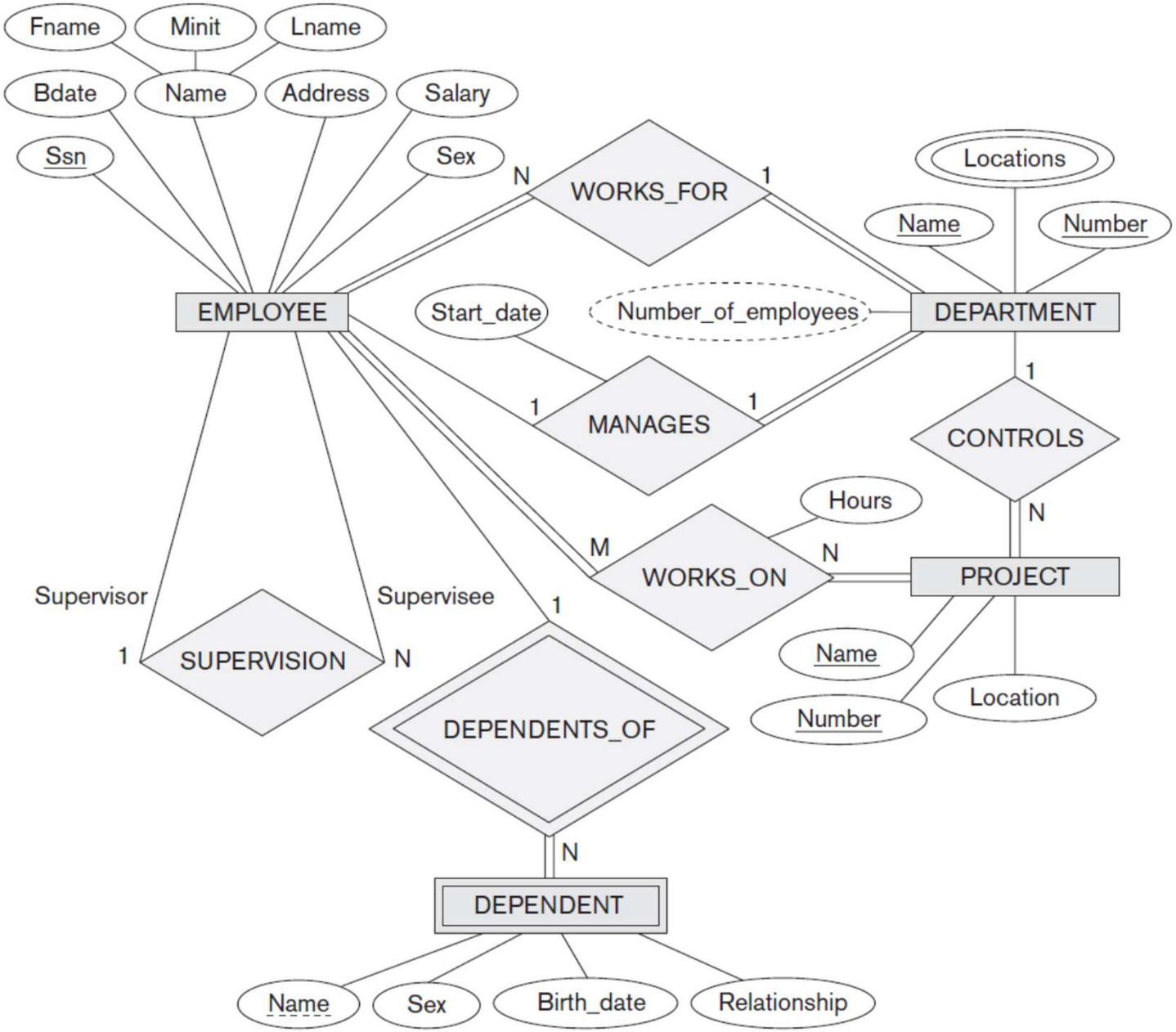
COMPANY Database

- Requirements (cont'd)
  - Employee: store each employee's name, Social Security number, address, salary, sex (gender), and birth date
  - Employee is assigned to one department
  - Employee may work on several projects
  - Keep track of number of hours per week that employee work on each project
  - Keep track of the direct supervisor

COMPANY Database

- Requirements (cont'd)
  - Keep track of the dependents of each employee
  - Dependent: dependent's first name, sex, birth date, and relationship to the employee

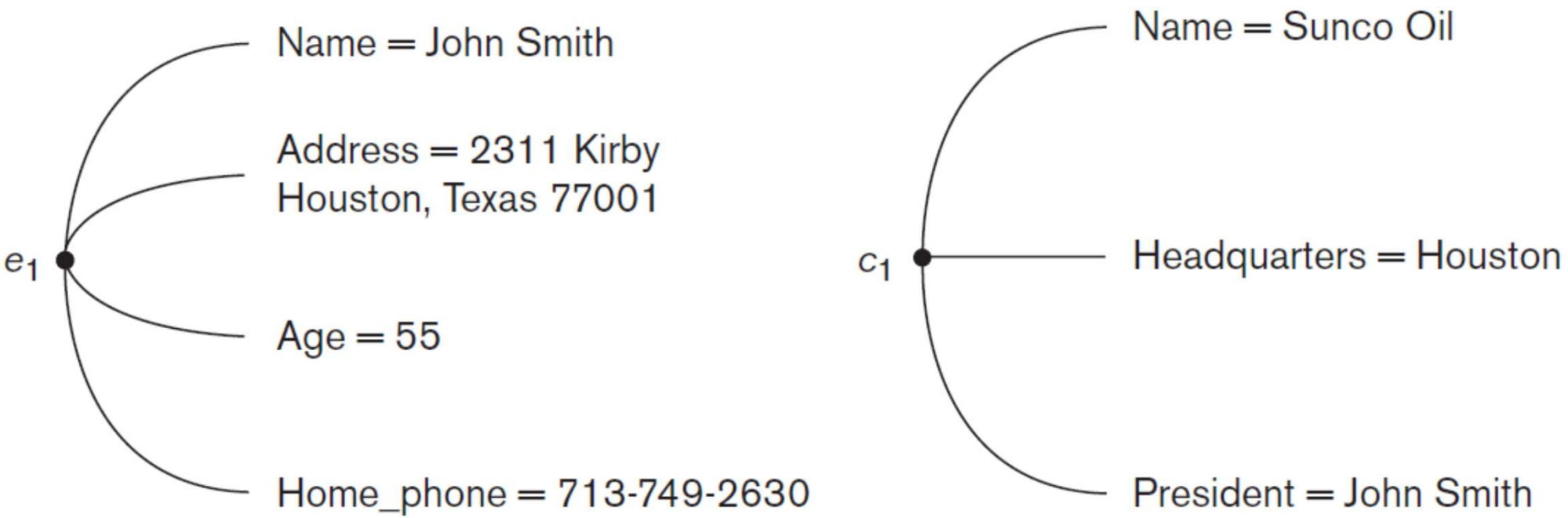
COMPANY Database



Entities and Attributes

- Entity
  - ***Thing*** in real world with independent existence
  - Examples: person, car, house, student, employee, …
- Attributes
  - Particular ***properties*** that describe the entity
  - Examples: name, age, address, job, phone number, …

Entities and Attributes

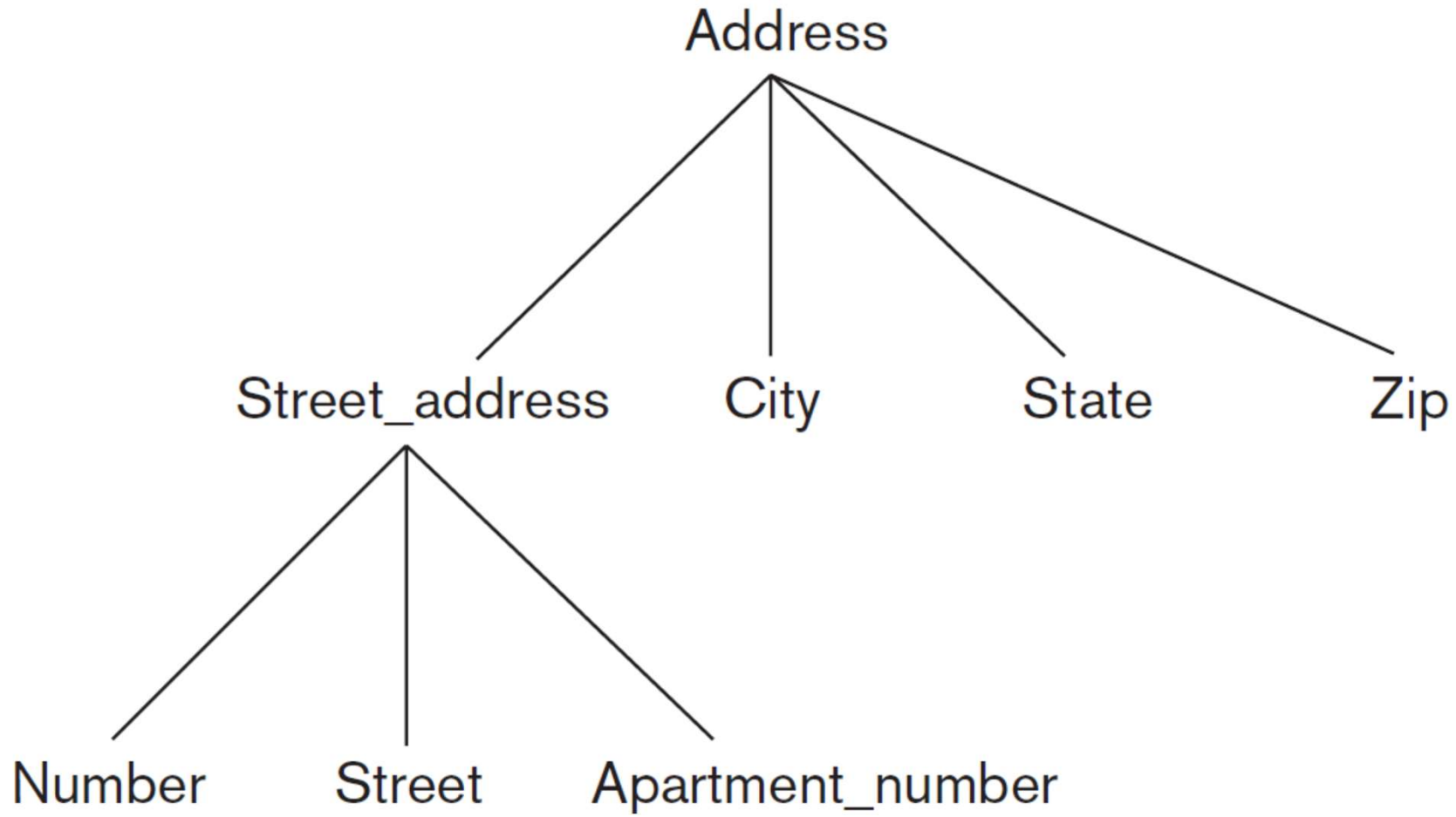


Types of Attributes

- Category 1
  - Simple (atomic) attributes
    - Attributes that are ***not divisible***
  - Composite attributes
    - Can be ***divided into smaller subparts***
    - Can form a ***hierarchy***

Types of Attributes

- A hierarchy of composite attributes



Types of Attributes

- Category 2
  - Single-valued attributes
    - Have a ***single value*** for a particular entity
    - Example: age
  - Multivalued attributes (set attributes)
    - Have a ***set of values*** for the same entity
    - Example: phone numbers

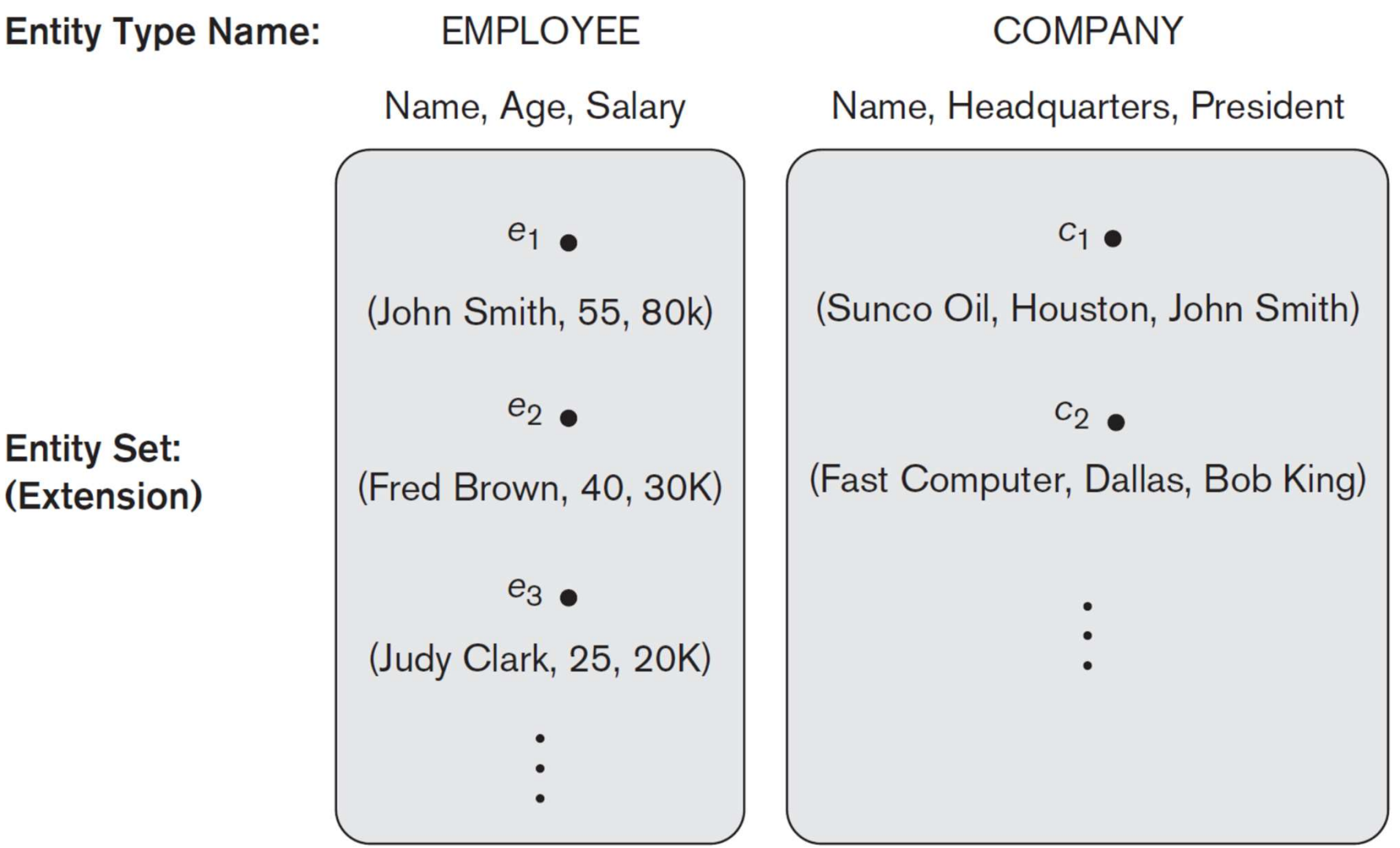
Types of Attributes

- Category 3
  - Stored attributes
    - Actually ***stored*** in the database
    - Example: birth date
  - Derived attributes
    - Can be ***derived*** from other attributes
    - Not stored in the database
    - Example: age (can be derived from birth date)

Entity Type

- Defines a collection of entities that have a ***same set of attributes***
- Indicates the ***schema*** or ***intention*** for a set of entities
- Described by its ***name*** and ***attributes***

Entity Type


Entity Type Name:
EMPLOYEE
Name, Age, Salary
COMPANY
Name, Headquarters, President
Entity Set:
(Extension)
e1
(John Smith, 55, 80k)
e2
(Fred Brown, 40, 30K)
e3
(Judy Clark, 25, 20K)
c1
(Sunco Oil, Houston, John Smith)
c2
(Fast Computer, Dallas, Bob King)

Entity Type

- Domains
  - Set of values that may be assigned to the attribute for each individual entity
  - Example
    - Domain for AGE attribute of EMPLOYEE entity: 20 ~ 70

Key Attribute

- Attributes whose values are ***distinct*** for each individual entity
- Values of key attributes can be used to ***identify*** each entity ***uniquely***
- Example: student number of STUDENT entity

Key Attribute

- Multiple attributes can form a key
- A entity type can have *more than two keys*
- Preceding uniqueness property of a key must hold for ***every entity set*** of the entity type
  - Not for a particular entity set

Key Attribute

CAR

Registration (Number, State), Vehicle_id, Make, Model, Year, {Color}

CAR_1

((ABC 123, TEXAS), TK629, Ford Mustang, convertible, 2004 {red, black})

CAR_2

((ABC 123, NEW YORK), WP9872, Nissan Maxima, 4-door, 2005, {blue})

CAR_3

((VSY 720, TEXAS), TD729, Chrysler LeBaron, 4-door, 2002, {white, blue})

Initial Conceptual Design of the COMPANY Database

DEPARTMENT
Name, Number, {Locations}, Manager, ManagerStartDate

PROJECT
Name, Number, Location, ControllingDepartment

EMPLOYEE
Name(FName, MInit, LName), SSN, Sex, Address, Salary,
BirthDate, Department, Supervisor, {WorksOn (Project, Hours)}

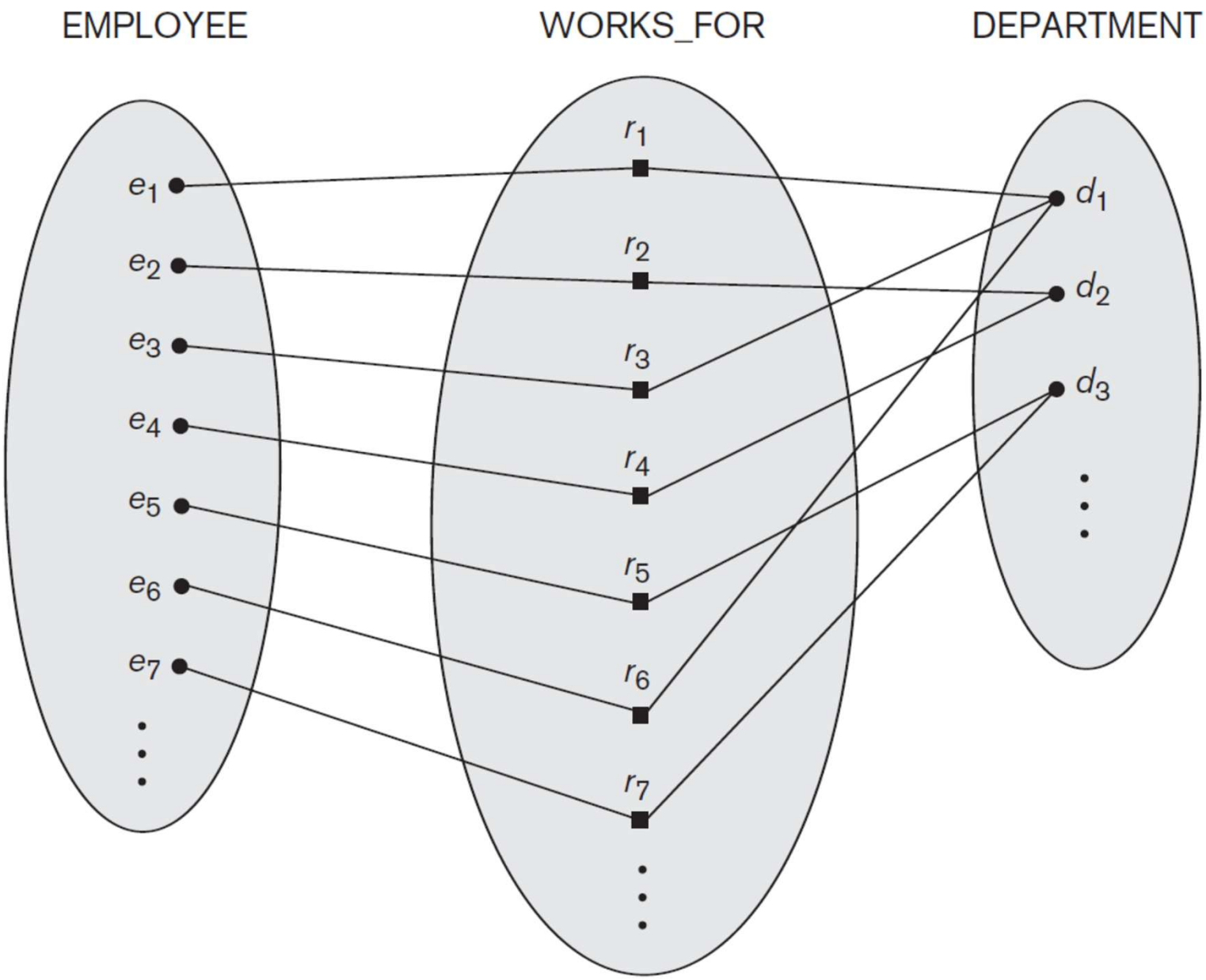
DEPENDENT
Employee, DependentName, Sex, BirthDate, Relationship

{}: multivalued attribute, (): composite attribute

Concepts of Relationship

- Relationship instances
  - ***Association*** among entities from entity sets
- Relationship type
  - Indicates ***a set of relationship instances*** from entity types
  - Represented with ***participating*** entity types

WORKS_FOR Relationship



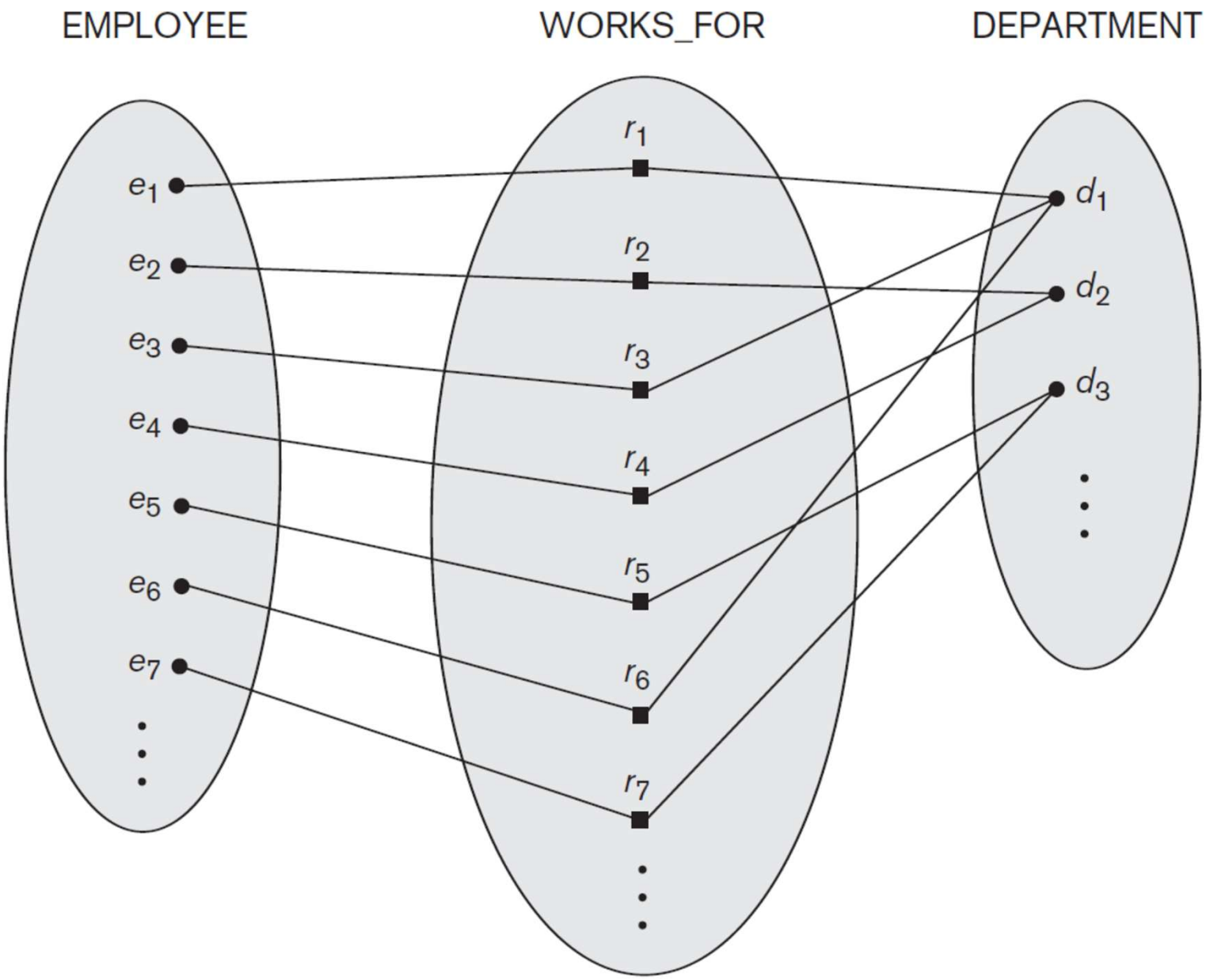
Relationship Degree

- ***Degree*** of a relationship type
  - Number of participating entity types
  - ***Binary, ternary, ...***

Binary Relationship

- A relationship type of ***degree two***
- Most commonly used
- Example
  - WORKS_FOR relation type has EMPLOYEE and DEPARTMENT as participating entity types

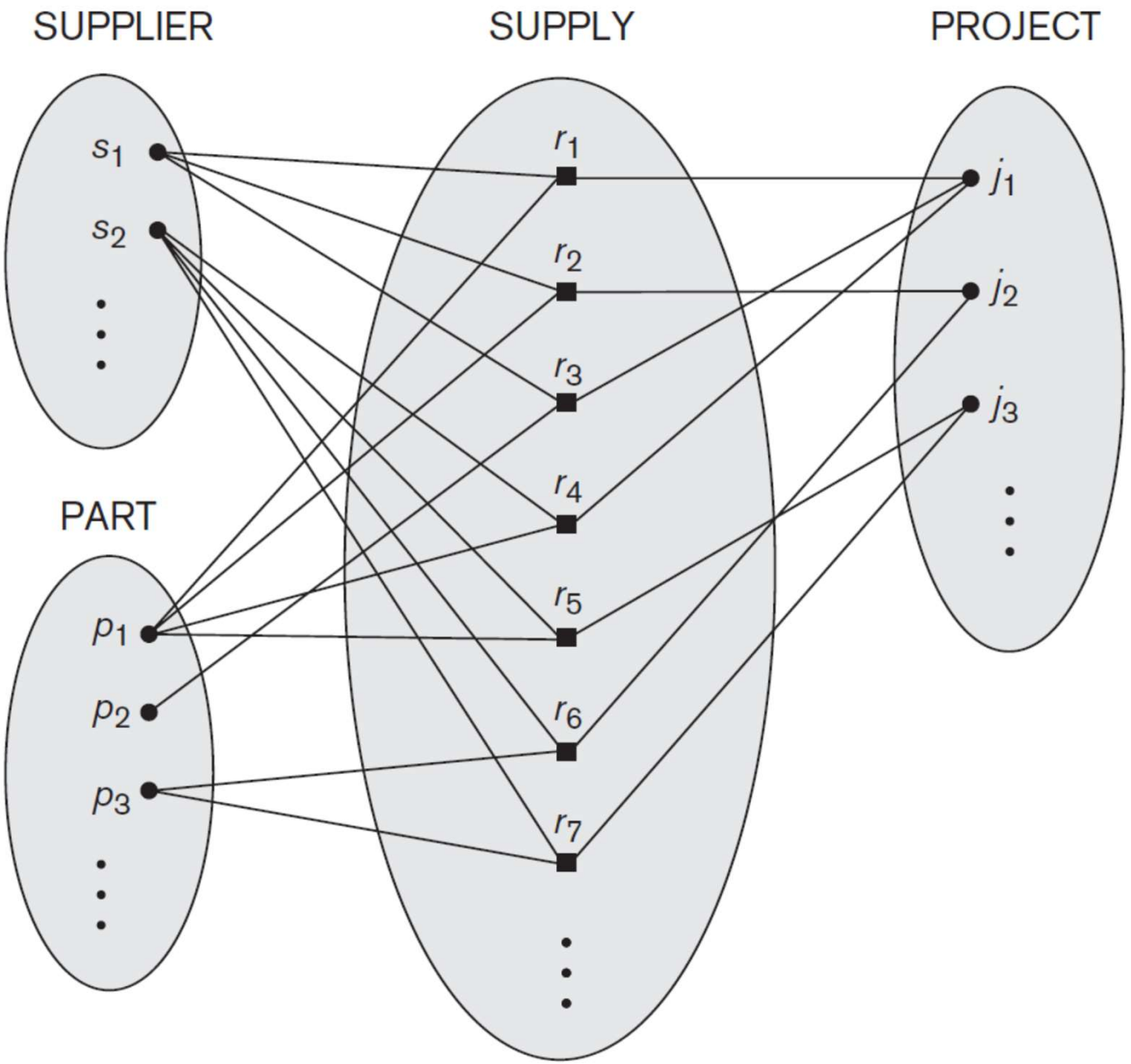
WORKS_FOR Relationship



Ternary Relationship

- A relationship type of ***degree three***
- Example
  - SUPPLY relation type has SUPPLIER, PART, and PROJECT as participating entity types

SUPPLY Relationship



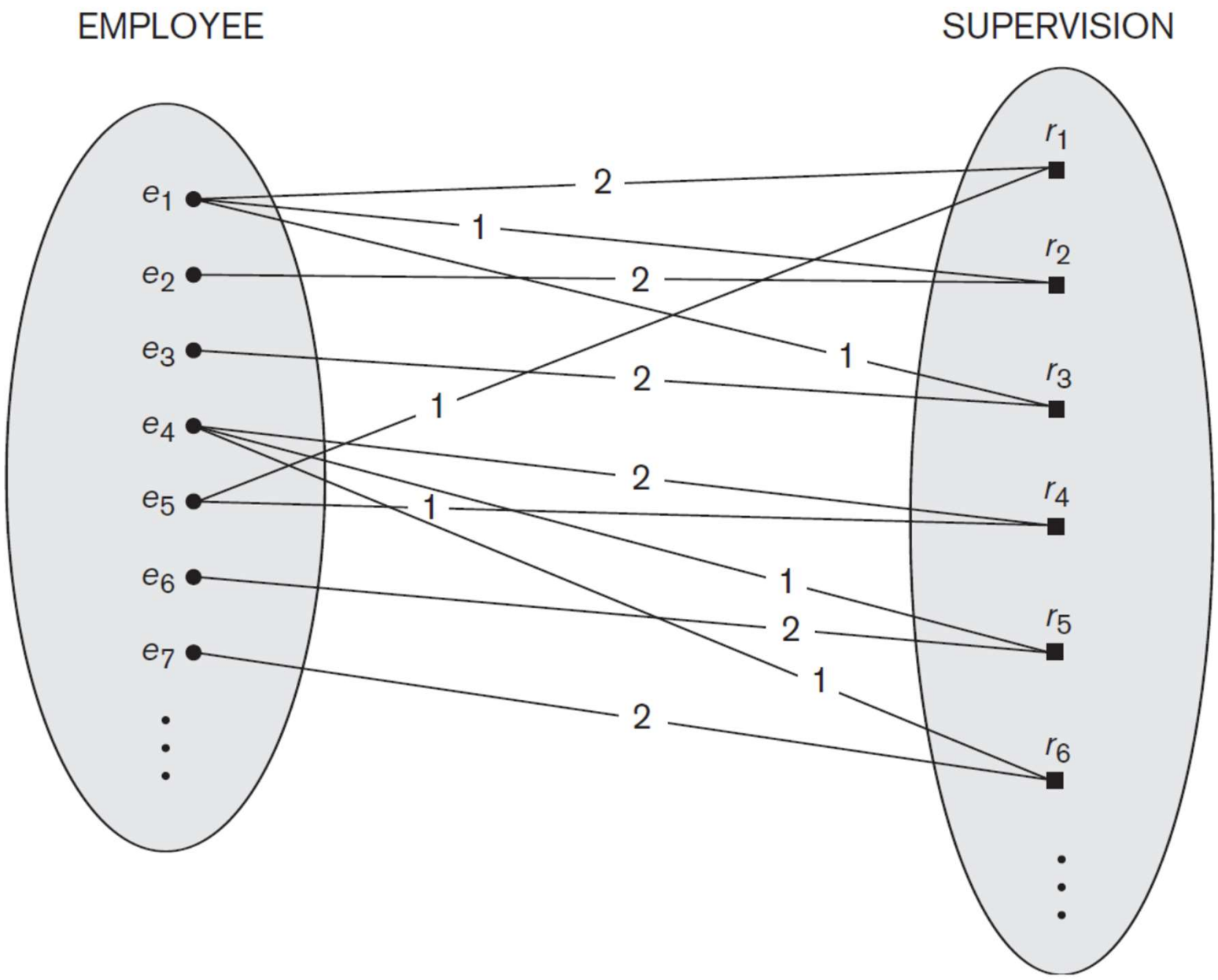
Recursive Relationships

- Role name
  - Signifies a role that a participating entity plays in each relationship instance
  - Example: EMPLOYEE and DEPARTMENT in WORKS_FOR relation type

Recursive Relationships

- Recursive relationship type
  - The same entity type participates more than once in a relationship type in different roles
  - Must specify a role name

Recursive Relationships



Summary

- Database design process
- Basic ER model concepts of entities and their attributes
  - Different types of attributes
  - Structural constraints on relationships

References

1. Batini, Carlo., S. Ceri, S. B. Navathe, and Carol Batini, *Conceptual Database Design: an Entity/Relationship Approach*, Addison-Wesley, Reading MA, 1991.
2. Chen, P. P., "The entity-relationship model: toward a unified view of data," *ACM Trans. on Database Systems* **1**:1, pp. 9-36, 1976.
3. Thalheim, B., *Fundamentals of Entity-Relationship Modeling*, Springer-Verlag, Berlin, 2000.

Have a nice day!